

Numerical Analysis PhD Exam, May 2006

Do any eight of the ten problems below.

1. Let $x_0, \dots, x_n$ be distinct real points. Let $\{l_j(x)\}_{j=0, \dots, n}$ be the Lagrange basis functions for these points. Prove that

$$\sum_{j=0}^n l_j(x) = 1$$

for all x .

2. Suppose you want to solve numerically the ode

$$y' = f(x, y), \quad y(0) = y_0.$$

- (a) By applying the midpoint rule to the integral in the equation

$$y(x_{n+1}) = y(x_{n-1}) + \int_{x_{n-1}}^{x_{n+1}} f(s, y(s)) ds,$$

derive the midpoint method for odes.

- (b) Find the order of the local truncation error for this method.
- (c) Determine if this method is strongly stable, weakly stable or unstable.

3. Given x_0, x_1 and smooth $f(x)$, give a divided difference type formula for the cubic polynomial which satisfies:

$$\begin{aligned} p(x_0) &= f(x_0), \\ p'(x_0) &= f'(x_0), \\ p''(x_0) &= f''(x_0), \\ p(x_1) &= f(x_1). \end{aligned}$$

Also give a divided difference type error formula for $f(x) - p(x)$.

4. This problem has two parts.

- (a) Find the linear least squares approximation to $f(x) = x^2$ on the interval $[0, 2]$ by optimizing the least squares error over a and b in $r^*(x) = ax + b$.
- (b) Find the first two orthonormal polynomials $\phi_0(x), \phi_1(x)$ on $[0, 2]$ (weight function $w(x) = 1$). Use these to find the linear least squares approx to $f(x) = x^2$, and check this result agrees with the previous problem.

5. Consider the root-finding iteration defined by

$$x_{n+1} = x_n - f(x_n) \left[\frac{f(x_n)}{f(x_n + f(x_n)) - f(x_n)} \right].$$

This is called Steffensen's method. Show that the method converges quadratically when applied to $f(x) = x^2 - a$. (You may assume x_0 is close enough to the root.)

6. This problem has three parts.

- (a) Evaluate the p -norm of a diagonal matrix, $1 \leq p \leq \infty$.
- (b) Evaluate the p -norm of a rank one matrix $\mathbf{u}\mathbf{v}^*$, $1 \leq p \leq \infty$, $\mathbf{u} \in \mathbb{C}^m$ and $\mathbf{v} \in \mathbb{C}^n$.
- (c) Suppose $\mathbf{A} \in \mathbb{C}^{m \times n}$ can be expressed

$$\mathbf{A} = \sum_{k=1}^r \sigma_k \mathbf{u}_k \mathbf{v}_k^*,$$

where the vectors $\{\mathbf{u}_k : 1 \leq k \leq r\}$ and $\{\mathbf{v}_k : 1 \leq k \leq r\}$ are orthonormal. What is $\|\mathbf{A}\|_2$?

7. This problem has two parts.

- (a) Given $\mathbf{A} \in \mathbb{C}^{m \times n}$, show that $\mathbf{A}^* \mathbf{A}$ is positive semidefinite and $\mathbf{A}^* \mathbf{A}$ is positive definite if and only if the columns of $\mathbf{A}$ are linearly independent.
- (b) If $\mathbf{A}$ is a Hermitian, positive semidefinite matrix, show that the diagonal contains the largest in magnitude element of $\mathbf{A}$. Hint: show that $|a_{ij}| \leq \max\{|a_{ii}|, |a_{jj}|\}$.

8. Let $\mathbf{A} \in \mathbb{C}^{n \times n}$ be nonsingular. Show that $\mathbf{A}$ has an LU factorization if and only if for each k with $1 \leq k \leq n$, the upper-left $k \times k$ block $\mathbf{A}_{1:k, 1:k}$ is nonsingular. Prove that this LU factorization is unique.

9. Consider the Arnoldi iteration on the Krylov spaces $\mathcal{K}_k = \text{span}\{\mathbf{b}, \mathbf{A}\mathbf{b}, \mathbf{A}^2\mathbf{b}, \dots, \mathbf{A}^{k-1}\mathbf{b}\}$ for some $\mathbf{b} \in \mathbb{C}^m$ and $\mathbf{A} \in \mathbb{C}^{m \times m}$ (given at side). Suppose the algorithm proceeds without breakdown until for some $n < m$, it encounters $h_{n+1, n} = 0$.

Algorithm 1 (Arnoldi iteration)

(a) Set $\mathbf{q}_1 = \mathbf{b}/\|\mathbf{b}\|_2$.

(b) For $n = 1, 2, 3, \dots$ do:

i. Set $\mathbf{v} = \mathbf{A}\mathbf{q}_n$.

ii. For $j = 1, 2, \dots, n$ do:

A. Set $h_{jn} = \mathbf{q}_j^* \mathbf{v}$.

B. Replace $\mathbf{v}$ by $\mathbf{v} - h_{jn} \mathbf{q}_j$.

iii. Set $h_{n+1, n} = \|\mathbf{v}\|_2$.

iv. Set $\mathbf{q}_{n+1} = \mathbf{v}/h_{n+1, n}$.

(a) Show that $\mathcal{K}_n$ is an invariant subspace of $\mathbf{A}$, i.e., $\mathbf{A}\mathcal{K}_n \subseteq \mathcal{K}_n$.

(b) Show that $\mathcal{K}_n = \mathcal{K}_{n+1} = \mathcal{K}_{n+2} = \dots$.

(c) Let $\mathbf{H}_n$ be the $n \times n$ Hessenberg matrix whose ij -th entry ($i \leq j+1$) is the number h_{ij} computed in the algorithm. Show that each eigenvalue of $\mathbf{H}_n$ is an eigenvalue of $\mathbf{A}$.

10. Consider an invertible linear system $\mathbf{A}\mathbf{x} = \mathbf{b}$ and a perturbation $(\mathbf{A} + \delta\mathbf{A})(\mathbf{x} + \delta\mathbf{x}) = \mathbf{b}$. Assuming $\|\mathbf{A}^{-1}\| \|\delta\mathbf{A}\| < 1$, derive the condition estimate

$$\frac{\|\delta\mathbf{x}\|}{\|\mathbf{x}\|} \leq \left(\frac{\|\mathbf{A}\| \|\mathbf{A}^{-1}\|}{1 - \|\mathbf{A}^{-1}\| \|\delta\mathbf{A}\|} \right) \frac{\|\delta\mathbf{A}\|}{\|\mathbf{A}\|}.$$

For given $\mathbf{A}$ and $\mathbf{x}$ and for the 2-norm, what choice of $\delta\mathbf{A}$ yields the following equality:

$$\|\mathbf{A}^{-1} \delta\mathbf{A} \mathbf{x}\| = \|\mathbf{A}^{-1}\| \|\delta\mathbf{A}\| \|\mathbf{x}\|?$$